

Lecture 16: Segmentation

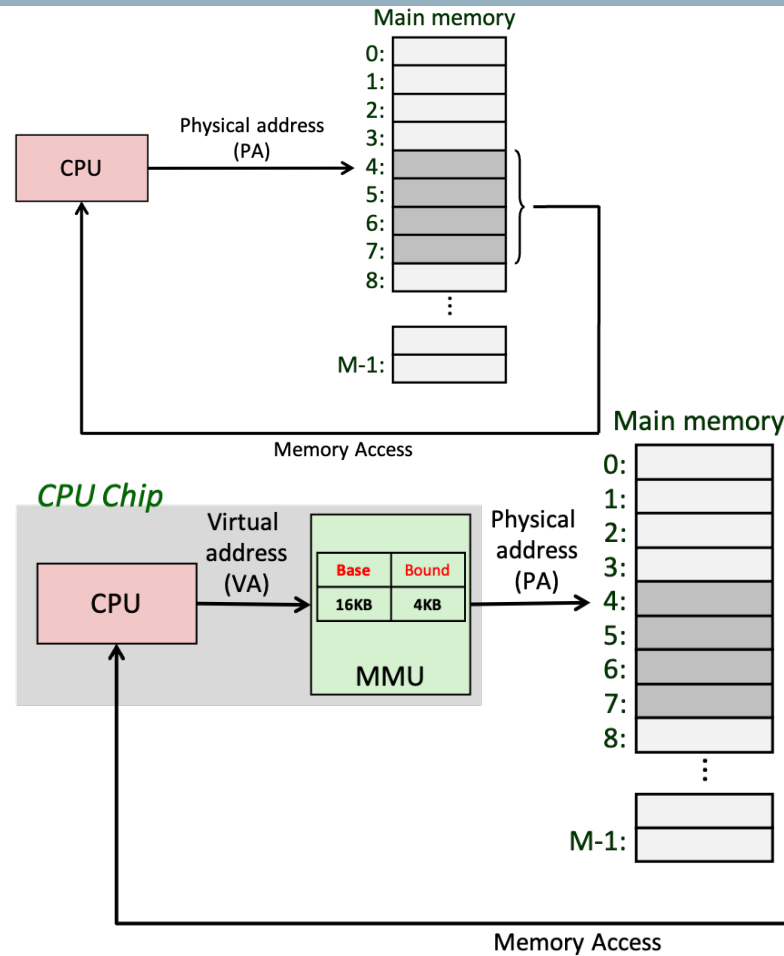
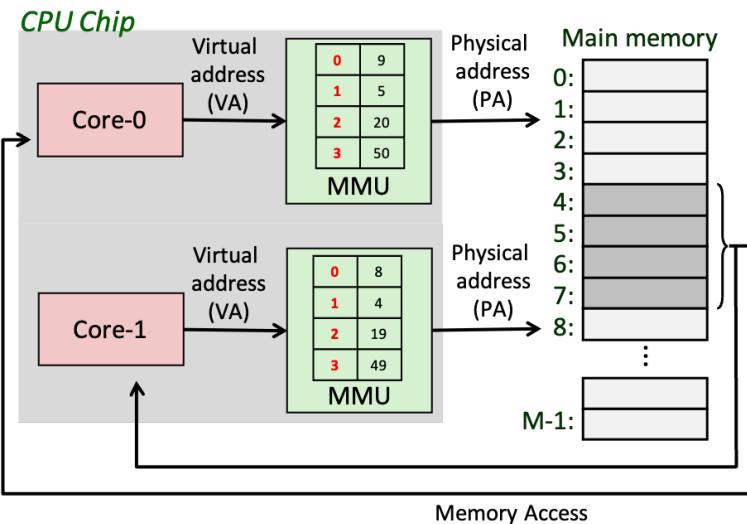
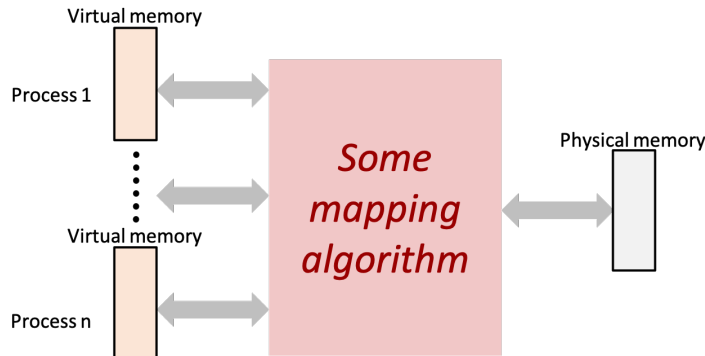
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Last Lecture



Virtual Address		Physical Address
0	→	16 KB
1 KB	→	17 KB
3000	→	19384
4400	→	Fault (out of bounds)

- We need some kind of mapper between VM-PM as VM (virtual memory) too big than available PM (physical memory)
 - One-to-one mapping
 - Using MMU table
 - Dynamic relocation using base/bound registers
 - Prone to fragmentation

Today's Class

- Segmentation
- Segment registers
- Global Descriptor Table (GDT)
- Quiz-3

Loader View

```
vivek@possum:~/os23$ readelf -l fib

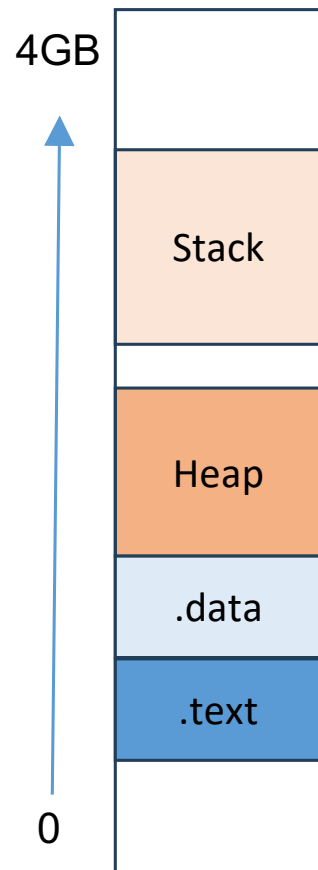
Elf file type is DYN (Position-Independent Executable file)
Entry point 0x1080
There are 11 program headers, starting at offset 52

Program Headers:
Type           Offset   VirtAddr   PhysAddr   FileSiz MemSiz  Flg Align
PHDR           0x000034 0x00000034 0x00000034 0x00160 0x00160 R   0x4
INTERP        0x000194 0x00000194 0x00000194 0x00013 0x00013 R   0x1
    [Requesting program interpreter: /lib/ld-linux.so.2]
LOAD          0x000000 0x00000000 0x00000000 0x003f8 0x003f8 R   0x1000
LOAD          0x001000 0x00001000 0x00001000 0x00278 0x00278 R E 0x1000
LOAD          0x002000 0x00002000 0x00002000 0x00134 0x00134 R   0x1000
LOAD          0x002ed4 0x00003ed4 0x00003ed4 0x00134 0x0013c RW 0x1000
DYNAMIC       0x002edc 0x00003edc 0x00003edc 0x000f8 0x000f8 RW 0x4
NOTE         0x0001a8 0x000001a8 0x000001a8 0x00044 0x00044 R   0x4
GNU_EH_FRAME 0x002020 0x00002020 0x00002020 0x0003c 0x0003c R   0x4
GNU_STACK    0x000000 0x00000000 0x00000000 0x00000 0x00000 RW 0x10
GNU_RELRO    0x002ed4 0x00003ed4 0x00003ed4 0x0012c 0x0012c R   0x1

Section to Segment mapping:
Segment Sections...
00
01  .interp
02  .interp .note.gnu.build-id .note.ABI-tag .gnu.hash .dynsym .dynst
03  .init .plt .plt.got .text .fini
04  .rodata .eh_frame_hdr .eh_frame
05  .init_array .fini_array .dynamic .got .data .bss
06  .dynamic
07  .note.gnu.build-id .note.ABI-tag
08  .eh_frame_hdr
09
10  .init_array .fini_array .dynamic .got
```

- PHDR index 3
 - Starting virtual address is 4096 and total size 632
- PHDR index 4
 - Starting virtual address is 8192 and total size 308
- PHDR index 5
 - Starting virtual address is 16084
- Non-contiguous memory layout across segments (inter-segment)
- Contiguous memory within segments (intra-segment)

ELF Segment Allocations



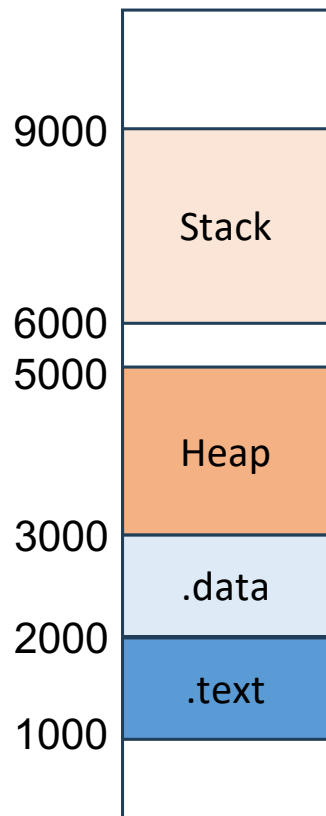
- Exact size for stack and heap size cannot be known upfront
 - Hello world program
 - Minimal sized stack and heap
 - Recursive Fibonacci program
 - Large stack but minimal sized heap
 - Recursive merge sort program
 - Large stack and large heap
- Observations
 - Each segment (stack, heap, data and text) has different memory requirements

Mapping Virtual to Physical Address

- Approaches

1. Virtual address is same as the physical address
2. Virtual address is not the same as physical address
 - a) Using base & bound for VA to PA mapping
 - b) Using segmentation for VA to PA mapping

Approach-2: Segment Wise Base/Bound

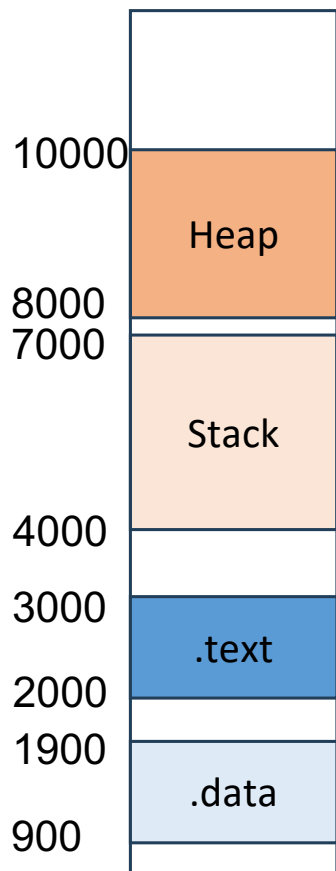


- Each segment can have its own (“personal”) base and bound values
 - **Segmentation**
- Each virtual address can have few bits to specify the segment they belong (e.g., 2 bits to specify 4 different segments), and remaining bits to specify the offset into that segment



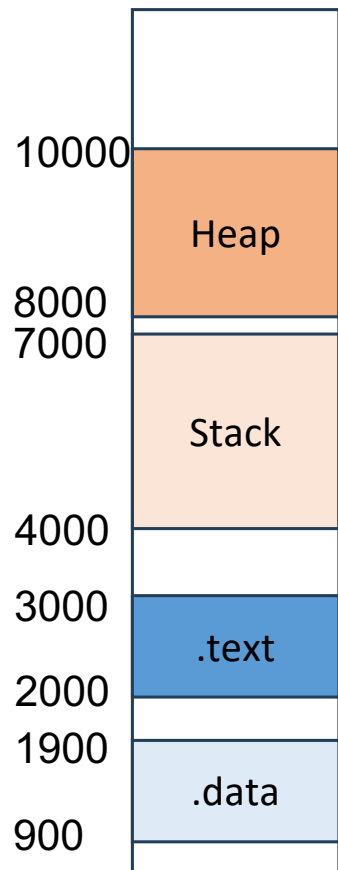
- Example: How to access virtual address 250 in .data (binary: **1111010**)?
 1. Find the segment from the segment selector (e.g., data for first two bits)
 2. Find the offset (111010) = 58
 3. Find the **base** for this segment (Base=2000)
 4. Check if offset within **bounds** (Bound=1000)
 5. Physical address = 2058

Non-Contiguous Inter-Segment Memory



- As the base and bound is separate for each segment, now it is not a requirement to have a contiguous chunk of physical address space for the process's address space
 - Better space management and reduction in fragmentation
- Example: How to access virtual address 250 in .text (binary: **10111010**)?
 1. Find the segment from the segment selector (e.g., data for first two bits)
 2. Find the offset (111010) = 58
 3. Find the **base** for this segment (Base=900)
 4. Check if offset within **bounds** (Bound=1000)
 5. Physical address = **958**

Accessing Segment's Base/Bound

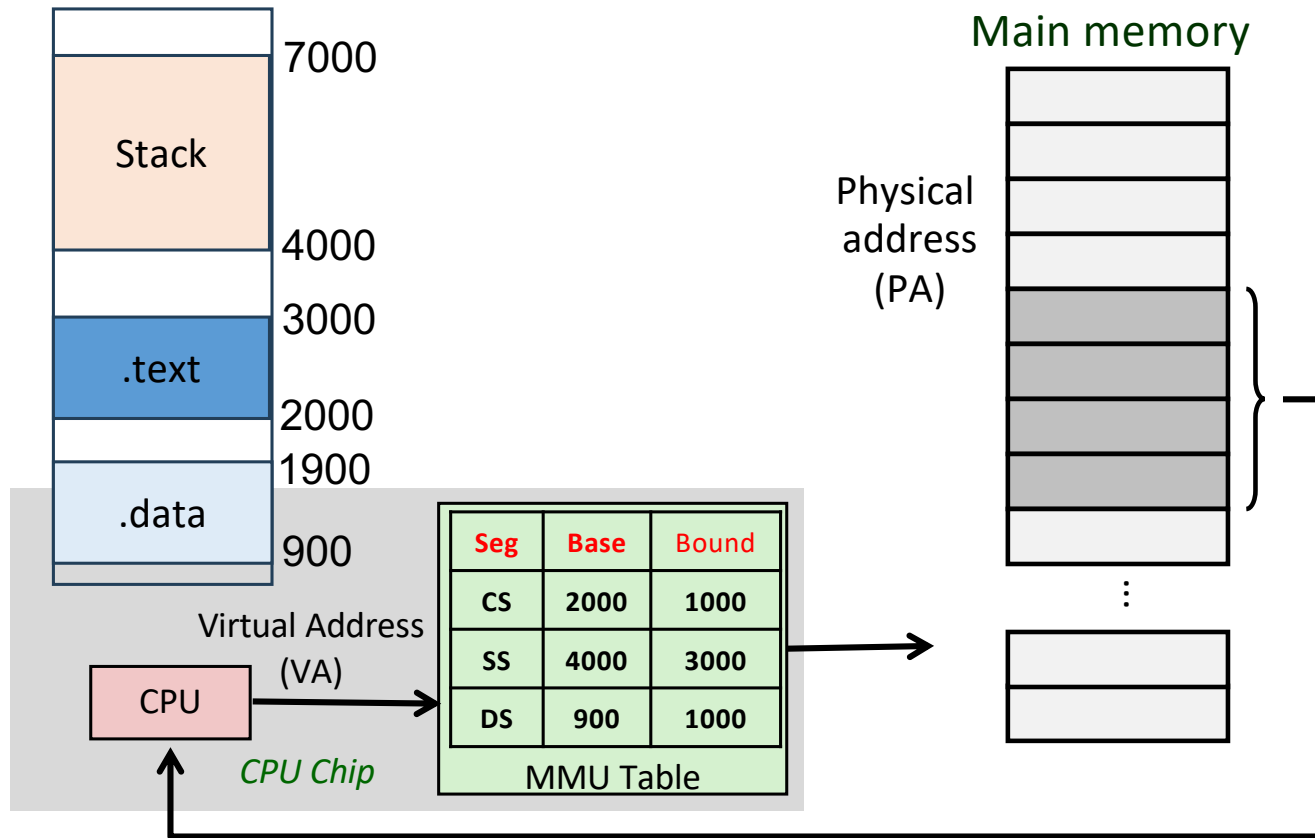


- How to access base and bounds of individual segments?
- Can we store the segments and their base/bound values in a MMU table?
 - Yes, but it would require **iterating** through the MMU table entries [$O(n)$]
 - **Any simple technique?**
 - Hardware support by providing dedicated registers on each core

Hardware Support for Segmentation

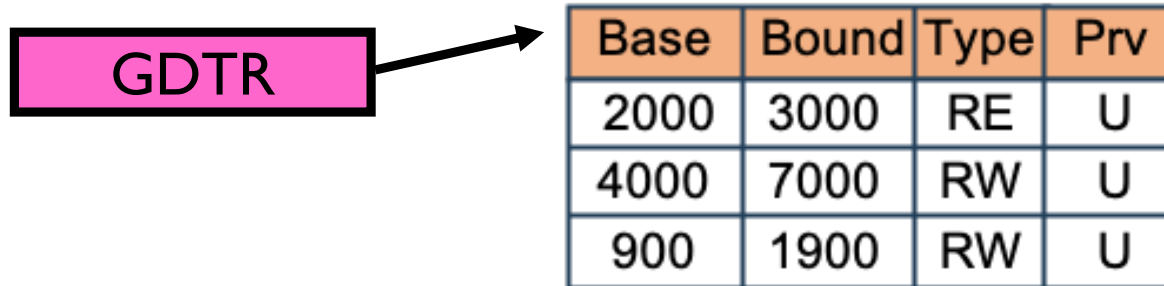
- x86 platform supports six segment registers per CPU
 - **CS (Code Segment)**
 - **SS (Stack Segment)**
 - **DS (Data Segment)**
 - ES (e.g., string operations), FS and GS for thread/CPU local operations
 - General purpose
- Reduced address translation time as single CPU cycle required to access registers
- A program could have several segments, but at any given time it can only use a maximum of 6 segment registers on x86
- To simplify the understanding, let us assume instead of base/bound register, now each CPU just have segment registers

Segment Registers



- MMU table can now have entries for each segment registers and their corresponding base and bound values
- **How to have low latency access to MMU table base address?**

Global Descriptor Table (GDT)



- Similar to IDT, there is a GDT (our MMU table) stored in memory (DRAM)
 - Usually one GDT per processor
- Each CPU has a dedicated register (GDTR) that holds the starting address of this GDT
- Each entries in GDT has a base-bound pair, privilege mode for this physical memory segment (kernel vs user space), memory access type (read/write/executable), etc.
- Still, too many operations for each virtual memory accesses
 1. Load segment selector from the virtual address
 2. Load GDT **offset** by looking into segment register (at the start, the segment register will require loading)
 3. Load GDTR and add the above **offset** to get index
 4. Load base and bound values stored at that index
- How to directly access segment's base/bound?
 - Add hardware support for base/bound per segment register!

Note: There is also another table Local Descriptor Table (LDT) in x86, which is per-process unlike the GDT. GDT can store base/bound for the LDTs

Caching of Base/Bound Per Register

Visible Part	Hidden Part	
GDT index	Base Address, Limit, Access Information	CS
		SS
		DS
		ES
		FS
		GS

- Each segment register has a visible and a hidden part
 - Visible part is programable and it is an index into the GDT for that particular segment register
 - The hidden part is non-programable and it stores the base, bound (a.k.a limit) and access information for each segment register (cache)
 - It is cached automatically the first time a GDT index is accessed
 - No need to access GDT until a context switch is done (**Why?**)
 - No extra CPU cycles wasted for accessing GDT in VA to PA translation

A System Using Segmentation

```
void scheduler() {
    while(true) {
        lock(process_table);
        foreach(Process p: scheduling_algorithm(process_table)) {
            if(p->state != READY) {
                continue;
            }
            p->state = RUNNING;
            unlock(process_table);
            swtch_segment_registers(scheduler_process, p);
            swtch(scheduler_process, p);
            // p is done for now..
            lock(process_table);
        }
        unlock(process_table);
    }
}
```

- Reload the segment registers during context switch
- It also automatically updates the values stored inside the hidden part of each segment register (see previous slide)

Points to Ponder

- **Can user process update GDTR to point to a new GDT?**
 - No, it can only be done at the boot time by the OS running in privileged mode
- **As GDT is stored in memory, can some rouge process can easily change/access the values stored inside the GDT?**
 - No, GDT is stored in a location that is accessible only to the OS
 - Updates to GDT entries are only carried out by OS in privileged mode (e.g., allocating process address space during exec)
- **Can a user process update its segment register?**
 - Yes, there are only 6 segment registers but a process could have several segments, and it might want to switch to a different segment for which it will update its segment register

Segmentation in Linux

- Linux uses segmentation in a very limited form on modern x86 64-bit processors
 - Simplifies the memory management
 - Portability across non-x86 architectures
- Each segment register has exact same base and bound (effectively disabling segmentation)
 - Base is 0 and bound is $2^{32} - 1$
 - Privilege is set as per kernel or user accessible segment
- How is then VA to PA mapping carried out?
 - Paging!

Segmentation Summary

- Advantages

- Limited hardware support (GDTR, 6 segments, etc.)
- Reduces internal fragmentation (exact amount of memory allocated)

- Disadvantages

- Difficult to increase the memory size once allocated if there is no contiguous chunk of unused physical memory of the new size
- Favors external fragmentation due to allocation of variable sized memory chunk for each segments
- Swapping out to disk has huge overheads, as an entire segment's memory would be swapped out if it is unused

Reference

- Chapter 16 in OSTEP book (“Segmentation”)

Next Lecture

- Introduction to paging
 - If you miss this lecture then you would have hard time understanding Lectures 18-20